

Original Hard-Packet Reconnection: Native-Atom Conservation, Proof-Carrying Reassembly, and an H8 Audit Certificate

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Abstract

TPC-MVP1 requires its packet architecture to return to the original TPC-15 scalar defect without changing the fixed shift, normalization, support, or provenance. We make that requirement machine-checkable. For

$$Q = \lfloor X^{1/2-\delta} \rfloor, \quad U = V = \lfloor Q^{1/2} \rfloor,$$

the original packet admits the exact opened form

$$B_{h_0, \delta}(X) = \sum_{\alpha=(\ell, k, d) \in \mathcal{A}_X} c_X(\alpha),$$

where

$$c_X(\ell, k, d) = -\Lambda(\ell)\mu(d)r_Q(\ell k + h_0)W(\ell k/X)$$

and $\mathcal{A}_X$ imposes $\ell > U$, $k > V$, $d \mid k$, $d \leq V$.

We encode a complete packet analysis by a canonical leaf matrix M_X^{can} , whose columns are native atoms and whose rows are final leaves. If $M_X = M_X^{\text{can}}$, if retained and soft leaves partition this matrix, and if

$$\mathbf{1}^T M_X = \mathbf{1}^T,$$

then

$$B_{h_0, \delta}(X) = \mathbf{1}^T M_X^{\text{ret}} c_X + \mathbf{1}^T M_X^{\text{soft}} c_X$$

exactly. A bound $\mathbf{1}^T M_X^{\text{soft}} c_X = o(X)$ then gives the required scalar reconnection. The certificate also requires the same h_0 , one physical normalization, equality of the actual and canonical leaf sets, and recoverable native keys. Merely observing the same scalar sum is weaker: it cannot detect null-pair inflation or equal-valued surrogate atoms.

Applied to the repository series through TPC-118 as audited on 26 July 2026, the result is a precise administrative gap statement. TPC-15 proves the scalar anchor; several later papers prove local identities, provenance interfaces, cover criteria and ledger rules. However, the audited snapshot supplies neither one complete canonical leaf archive from every opened TPC-15 atom through all retained and soft branches nor an alternative exact composite intertwining certificate. In particular, TPC-18 explicitly treats global block cover as an additional hypothesis. Hence the canonical-leaf H8 audit defined here is

not testable from the current artifacts.

This is a time-stamped gap certificate, not an impossibility theorem, not a claim that this archive format is logically necessary, and not an L2 arithmetic advance.

Keywords: fixed-shift hard packet; provenance; native atoms; exact reassembly; reconnection certificate.

1 Question, scope, and claim boundary

For fixed $h_0 \neq 0$, real $W \in C_c^\infty((0, \infty))$, and fixed $0 < \delta < 1/2$, TPC-15 proves

$$\Delta_{h_0, W}(X) = B_{h_0, \delta}(X) + O_{A, h_0, W, \delta}(X(\log X)^{-A}) \quad (1)$$

for every fixed $A > 0$ [13]. Thus $B_{h_0, \delta}(X) = o(X)$ is equivalent, inside that reduction, to the weighted Hardy–Littlewood asymptotic for an admissible even h_0 .

MVP1 Hypothesis H8 requires all later estimates and reassembly to intertwine with this same scalar packet. It forbids a second global normalization, an omitted branch, a duplicated atom, or a surrogate Gram matrix [1]. The present paper asks only whether that interface has a proof-carrying certificate. It does not estimate the packet.

Definition 1.1 (Verdict vocabulary). We use four route verdicts.

- (i) GO: every required input is proved in one quantifier scope;
- (ii) REROUTE: a specified subroute has a proved obstruction, but another route remains;
- (iii) INFEASIBLE: a theorem contradicts the target under its declared assumptions; and
- (iv) NOT TESTABLE: a required typed input or certificate is absent, with no contradiction proved.

The canonical-leaf H8 audit verdict below is NOT TESTABLE. Calling it INFEASIBLE would overstate the evidence; calling it GO would treat local identities as a global composite identity.

2 The TPC-15 native atom archive

Put

$$Q = \lfloor X^{1/2-\delta} \rfloor, \quad U = V = \lfloor Q^{1/2} \rfloor,$$

and

$$r_Q(n) = \Lambda(n) - \sum_{q \leq Q} \frac{\mu(q)}{\varphi(q)} c_q(n).$$

TPC-15 defines

$$B_{h_0, \delta}(X) = - \sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell) \beta_V(k) r_Q(\ell k + h_0) W(\ell k / X), \quad \beta_V(k) = \sum_{\substack{d | k \\ d \leq V}} \mu(d). \quad (2)$$

Definition 2.1 (Opened native atoms). Let $\mathcal{A}_X$ consist of triples

$$\alpha = (\ell, k, d)$$

satisfying

$$\ell > U, \quad k > V, \quad d \mid k, \quad d \leq V,$$

and for which $W(\ell k / X) \neq 0$. Define

$$c_X(\alpha) = -\Lambda(\ell) \mu(d) r_Q(\ell k + h_0) W(\ell k / X). \quad (3)$$

The key retains h_0, Q, U, V, W as global packet metadata.

Proposition 2.2 (Exact native-atom anchor). *One has the finite identity*

$$\boxed{B_{h_0, \delta}(X) = \sum_{\alpha \in \mathcal{A}_X} c_X(\alpha).} \quad (4)$$

Proof. Insert the divisor expansion of $\beta_V(k)$ in (2) and interchange finite sums. $\square$

Remark 2.3 (Why the opened key is necessary). The compressed pair (ℓ, k) is sufficient to evaluate the scalar packet, but it is not sufficient to audit later divisor-prefix, content, lcm, and determinant branches. Opening d makes the Möbius sign and every later partition assignment traceable. The opening is exact algebra, not an estimate.

3 A proof-carrying reconnection theorem

Let $\mathcal{L}_X$ be a finite set of final leaf records. A leaf may be a retained packet term or a term assigned to an exactly defined soft remainder. Linear analysis of the native coefficient vector

$$c_X = (c_X(\alpha))_{\alpha \in \mathcal{A}_X}$$

is described by a matrix

$$M_X \in \mathbb{C}^{\mathcal{L}_X \times \mathcal{A}_X}.$$

Its synthesized scalar is $\mathbf{1}^T M_X c_X$.

Definition 3.1 (Canonical leaf archive). A canonical leaf archive consists of:

- (a) an expected leaf set $\mathcal{L}_X^{\text{can}}$ generated by the declared partitions, polarizations, frequency bands, content branches, and boundary rules;
- (b) an expected matrix M_X^{can} ;
- (c) for every nonzero entry, a recoverable native atom, complete branch path, fixed-shift tag, and normalization tag; and
- (d) a disjoint typed partition

$$M_X^{\text{can}} = M_X^{\text{ret}} + M_X^{\text{soft}}.$$

An implementation passes the canonical audit only when its leaf set, keys, and matrix equal the declared archive entrywise.

Theorem 3.2 (Sufficient proof-carrying H8 certificate). *Assume:*

$$\mathcal{L}_X = \mathcal{L}_X^{\text{can}}, \quad M_X = M_X^{\text{can}}, \quad (5)$$

$$M_X = M_X^{\text{ret}} + M_X^{\text{soft}}, \quad \mathbf{1}^T M_X = \mathbf{1}^T, \quad (6)$$

and assume every active matrix entry retains the same prescribed h_0 , the TPC-15 normalization, and a recoverable $\alpha \in \mathcal{A}_X$. Then

$$\boxed{B_{h_0, \delta}(X) = \mathbf{1}^T M_X^{\text{ret}} c_X + E_{\text{soft}}(X)}, \quad E_{\text{soft}}(X) = \mathbf{1}^T M_X^{\text{soft}} c_X. \quad (7)$$

If, in addition, the complete outer soft sum satisfies

$$E_{\text{soft}}(X) = o(X),$$

then the retained synthesis reconnects to the original packet up to $o(X)$.

Proof. By [proposition 2.2](#) and (6),

$$B_{h_0, \delta}(X) = \mathbf{1}^T c_X = \mathbf{1}^T M_X c_X.$$

Insert the typed matrix partition in (6). The metadata hypotheses ensure that this equality concerns the same physical coefficient rather than an averaged shift or rescaled surrogate. $\square$

Remark 3.3 (Sufficient archive, not a unique proof language). The entrywise equality $M_X = M_X^{\text{can}}$ is a strong, machine-checkable sufficient certificate. MVP1 H8 could instead be established by another exact composite symbolic intertwining that proves the same no-omission, no-duplication, fixed-shift, normalization and provenance statements without materializing this particular matrix. Absence of the archive therefore does not prove that H8 is false or that no other proof exists.

Corollary 3.4 (Coefficientwise audit). *The conservation equation is equivalent to*

$$\sum_{\lambda \in \mathcal{L}_X} M_X(\lambda, \alpha) = 1 \quad (\alpha \in \mathcal{A}_X).$$

Thus deletion, uncompensated duplication, or global rescaling is detected column by column.

Remark 3.5 (Exact identity before estimation). A soft remainder is not permission to discard a branch. The branch first appears in the exact matrix partition; only then may its complete outer scalar be estimated by $o(X)$. Sectorwise or sampled smallness does not fill this slot.

4 Why scalar equality is not a provenance certificate

Proposition 4.1 (Null-pair and surrogate nonidentifiability). *Knowledge that*

$$\mathbf{1}^T M_X c_X = \mathbf{1}^T c_X$$

for one coefficient vector does not imply (5) or (6). This remains true even when the two scalar sums agree exactly.

Proof. Append to one atom two noncanonical leaves with multipliers t and $-t$. Their scalar contribution is zero, but the leaf archive has been inflated. Alternatively, if two native atoms happen to have the same numerical coefficient, exchange one for the other. The scalar does not change, while source provenance does. Both operations violate the canonical archive. $\square$

Example 4.2 (Second normalization). Replacing every leaf multiplier by $\gamma_X M_X$ gives a scalar $\gamma_X B_{h_0, \delta}(X)$. A later division by γ_X may restore the displayed number, but it introduces a second global normalization and changes every intermediate natural scale. H8 requires the normalization tag to remain the inherited physical one.

Example 4.3 (Shift substitution). An average

$$\frac{1}{|\mathcal{H}_X|} \sum_{h \in \mathcal{H}_X} B_{h, \delta}(X)$$

cannot be tagged h_0 merely because $h_0 \in \mathcal{H}_X$. The leaf archive must return the same prescribed shift term by term, unless a separate localization theorem is included with its cost.

Remark 4.4 (Quadratic surrogates). A TT^* , Gram, determinant-energy, or Fourier-energy identity is a tool for bounding a linear packet. It is not itself a reconnection map. The final archive must return from the quadratic statistic to the linear leaf coefficients with the exact prefactor and all losses recorded once.

5 Time-stamped artifact manifest and chain audit

The following manifest records the repository series through TPC-118 as inspected on 26 July 2026 and only the interfaces relevant to H8.

Source	Established interface	H8 consequence
TPC-15	Exact scalar reduction and the opened identity (4)	Native entrance is proved.
TPC-16	Exact square-root and dyadic normal forms on declared parameter ranges [14]	Local changes of variables are usable, but do not alone enumerate the later final leaves.
TPC-17	Exact dyadic partition and prefix–tail coefficient split on each admissible block [15]	The local column split is proved where its hypotheses hold.
TPC-18	Exact lcm, Gram, determinant, and content identities for symmetric tail blocks [16]	Its global closure corollary explicitly assumes a full block cover; that cover is not proved there.
TPC-93	Lossless local content extraction and affine export [17]	Supports a local provenance edge; it is not a composite archive from $\mathcal{A}_X$.
TPC-103	Six-ledger factor audit and one inherited normalization [2]	Confirms local atom metadata, not global completeness.
TPC-105	Recoverable provenance bag and input profile under equal-map quotienting [3]	Prevents local quotient loss; no all-branch leaf list is produced.
TPC-107	Exact finite reordering of the signed resonant sum [4]	A valid local linear identity, conditional on its imported atom family.
TPC-108–112	Exact arithmetic gates, Gram diagnostics, and exponent ledgers [5, 6]	These are estimates or criteria; they do not supply the missing composite reconnection matrix.
TPC-113–115	Quotient-frame conditioning, native residual grouping and the exact fixed-shift evaluation criterion [7–9]	Exact local operator criteria; the actual growing H9 composite, native census and physical shift profile remain uninstantiated.
TPC-116	Conditional complete-mask tail closure at the original normalization [10]	Defines the required archive and bounds, but does not verify the actual growing mask family or complete outer estimates.
TPC-117	Exact full-block range/residual criterion [11]	The actual growing B_X, w_X archive is absent, so the physical cover test is not executable in the snapshot.
TPC-118	Nonduplicated endpoint registry and exact certificate logic [12]	The actual theorem-backed registry is incomplete and does not supply a composite leaf archive.

Theorem 5.1 (Snapshot verdict for the canonical-leaf audit). *In the manifest above, the established artifacts do not contain the data required to evaluate (5)–(6) for the complete TPC-15 native archive, nor do they contain another complete exact intertwining certificate. Therefore*

$$\boxed{\text{Audit}_{\text{H8}}^{\text{can}} = \text{NOT TESTABLE FROM CURRENT ARTIFACTS.}}$$

Here “current” means the enumerated repository snapshot audited on 26 July 2026. This verdict neither proves that H8 is false nor supplies an L2 estimate.

Proof. The cited papers prove multiple local identities. However, no single artifact enumerates, for every $\alpha = (\ell, k, d) \in \mathcal{A}_X$, every final retained and soft descendant, its exact multiplier, branch path, shift tag, and normalization tag. In particular, the TPC-18 full-block return is stated under an additional global-cover hypothesis. Hence the actual matrix and canonical matrix required by (5) cannot presently be compared; the manifest also contains no alternative complete symbolic intertwining. $\square$

Remark 5.2 (Why this is not a global stop). The missing object is finite and algebraic at each X . It may be constructible from the existing local identities plus a complete cover theorem. No counterexample to that construction is proved here.

6 Minimal completion artifact and claim ledger

A future proof-carrying H8 artifact should contain:

- R1.** the exact TPC-15 metadata $(X, h_0, \delta, Q, U, V, W)$;
- R2.** every active opened atom (ℓ, k, d) and (3);
- R3.** the deterministic expected branch paths through every partition, polarization, content, frequency, boundary, and return rule;
- R4.** the actual final leaf set and exact multipliers;
- R5.** the retained/soft type of each leaf;
- R6.** column conservation, unique canonical leaf identifiers, and fixed-shift/normalization checks; and
- R7.** an analytic citation for the complete outer soft bound.

The accompanying script verifies the finite logic on an exact rational model. It rejects deletion, rescaling, a wrong shift, a second normalization, scalar-preserving null pairs, and equal-valued surrogate provenance. It does not pretend to contain the missing physical archive.

Statement	Status
TPC-15 opened native-atom identity	Exact L0/L1 interface.
Proof-carrying matrix criterion	Exact finite theorem (L0).
Existing local provenance transformations	Several proved L1 edges.
Complete physical leaf archive and column audit	Absent in the dated manifest; the canonical H8 audit is not testable.
Complete outer soft remainder $o(X)$	Must be cited branch by branch; not proved here.
Growing fixed- h_0 arithmetic saving	Not proved; no L2 event.

Explicit exclusions. This paper proves no signed affine Möbius cancellation, no determinant-energy lower bound, no zero-mode saving, no physical endpoint estimate, no parity breakthrough, no Hardy–Littlewood asymptotic, and no prime-pair or twin-prime theorem. Nothing is specialized to $h_0 = 2$.

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